Haiwei Yu

Applied AI / LLM Systems / AI Agents / AI Solutions

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Education

Duke University - M.S. in Computer Science

Expected May 2027 | GPA: 3.90/4.00

The University of Utah - B.S. in Computer Science

Jan 2021-May 2025 | GPA: 3.77/4.00 | Dean's List

Research Experience

Research Assistant - AI Systems for Health Data

Duke University | Jan 2026-Present

- Design and develop an LLM-based system that helps users query electronic health records in natural language, with validation and traceable query execution.
- Independently developed the system, conducted experiments and failure analysis, and prepared the first-author manuscript.

Selected Projects

UParkNow

The University of Utah · Capstone | Aug 2024-May 2025

Independent backend design and implementation; the full product was a team capstone

- Designed and implemented the Java Spring Boot backend, including REST APIs, authentication, data persistence, and mail integration.
- Built WebSocket communication to synchronize parking updates between the recognition system, Android clients, and the management interface.

Backend repository

PolyglotDRA-SecBench

Duke University | Mar-May 2026

- Built an offline English/Chinese benchmark for deep-research agents under clean and adversarial retrieval conditions.
- Integrated MiroThinker and Tongyi DeepResearch, normalized tool-use traces, and automated scoring of task utility and undesirable agent behavior, including goal drift, decoy engagement, and taint leakage.

SafeStart

Duke University | Sep 2026-Present

- Developing an AI assistant for mining pre-task planning that uses safety requirements, SOPs, and incident references to help draft job safety analyses, with supervisor review.

QueueHub (QHub)

Duke University | Aug 2025-May 2026

- Built a real-time TA queue with role-based access and atomic claim locking to coordinate concurrent teaching-assistant actions.

Internship Experience

Applied Machine Learning Intern

Institute of Automation, Chinese Academy of Sciences · Beijing | Jul-Aug 2023

- Built Python and OpenCV workflows for image collection, annotation, preprocessing, and quality checks.

Selected Publication

Leveraging Prompt Engineering to Facilitate AI-driven Chart Generation

Published | 2024 · CISAI 2024 · pp. 261-268 | DOI

Technical Skills

AI systems: Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), Agent Evaluation, Natural Language Processing (NLP), Text-to-SQL, Prompt Engineering

AI workflows & evaluation: Workflow Design, Human Review, Model Evaluation, Error Analysis

Programming: Python, Java, SQL, TypeScript / JavaScript, C++

ML & data: PyTorch, scikit-learn, Pandas, Polars, Spark / PySpark

Backend & application systems: Spring Boot, Spring Security, REST APIs, WebSocket, JPA / Hibernate

Delivery: Docker, Git / GitHub, CI/CD, AWS EC2 / RDS, Kubernetes (QueueHub)